



DATA SCIENCE

UNIT - III

Data Analytics Life Cycle

- Data Analytic Lifecycle: Introduction,
- Phase 1: Discovery,
- Phase 2: Data Preparation,
- Phase 3: Model Planning,
- Phase 4: Model Building,
- Phase 5: Communication results,
- Phase 6: Operationalize.

Data Analytics Lifecycle

- The Data Analytics Lifecycle is designed specifically for Big Data problems and data science projects.
- The lifecycle has six phases, and project work can occur in several phases at once.
- For most phases in the lifecycle, the movement can be either forward or backward.
- This iterative depiction of the lifecycle is intended to more closely describe a real project, in which aspects of the project move forward and may return to earlier stages as new information is uncovered and team members learn more about various stages of the project.
- This enables participants to move iteratively through the process and drive toward operationalizing the project work.

Key Roles for a Successful Analytics Project

■ **Business User:**

- Someone who understands the domain area and usually benefits from the results.
- This person can consult and advise the project team on the context of the project, the value of the results, and how the outputs will be operationalized.
- Usually a business analyst, line manager, or deep subject matter expert in the project domain fulfills this role.

■ **Project Sponsor:**

- Responsible for the genesis of the project.
- Provides the impetus and requirements for the project and defines the core business problem.
- Generally provides the funding and gauges the degree of value from the final outputs of the working team.
- This person sets the priorities for the project and clarifies the desired outputs.

Key Roles for a Successful Analytics Project

■ **Project Manager:**

- Ensures that key milestones and objectives are met on time and at the expected quality.

■ **Business Intelligence Analyst:**

- Provides business domain expertise based on a deep understanding of the data, key performance indicators (KPIs), key metrics, and business intelligence from a reporting perspective.
- Business Intelligence Analysts generally create dashboards and reports and have knowledge of the data feeds and sources.

Key Roles for a Successful Analytics Project

■ Database Administrator (DBA):

- Provisions and configures the database environment to support the analytics needs of the working team.
- These responsibilities may include providing access to key databases or tables and ensuring the appropriate security levels are in place related to the data repositories.

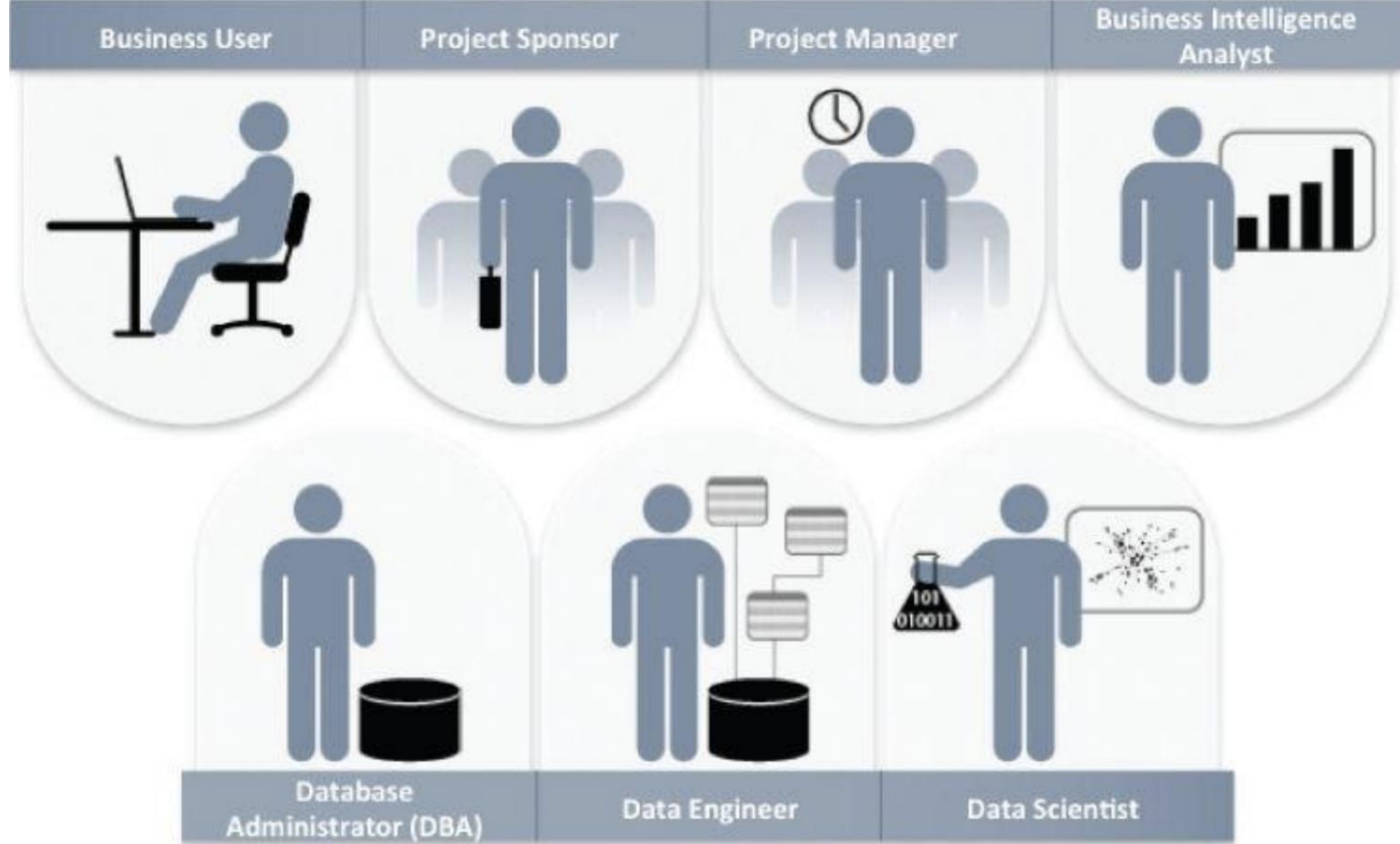
■ Data Engineer:

- Leverages deep technical skills to assist with tuning SQL queries for data management and data extraction, and provides support for data ingestion into the analytic sandbox.
- DBA sets up and configures the databases to be used, the data engineer executes the actual data extractions and performs substantial data manipulation to facilitate the analytics.
- The data engineer works closely with the data scientist to help shape data in the right ways for analyses

Key Roles for a Successful Analytics Project

■ Data Scientist:

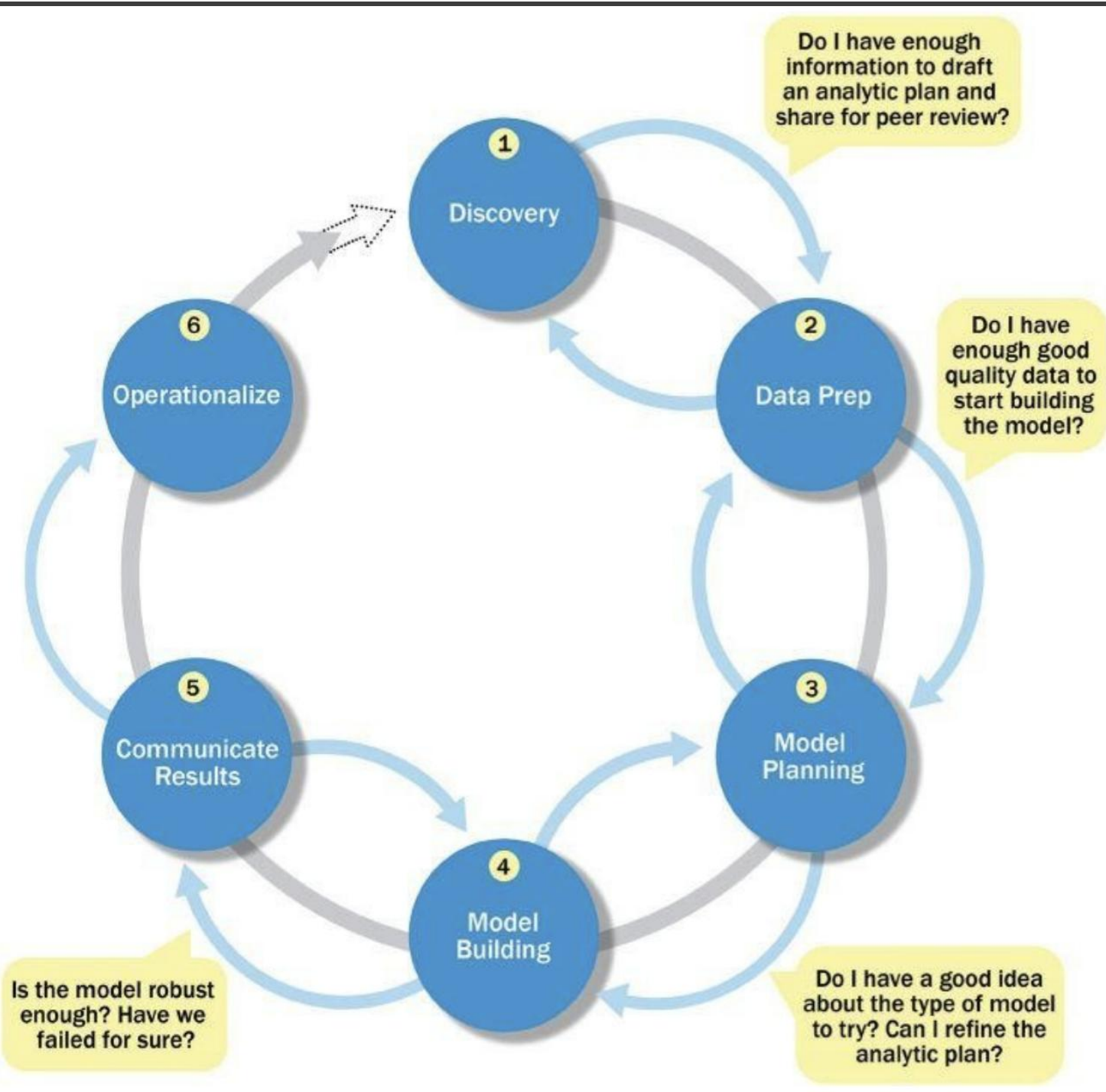
- Provides subject matter expertise for analytical techniques, data modeling, and applying valid analytical techniques to given business problems.
- Ensures overall analytics objectives are met.
- Designs and executes analytical methods and approaches with the data available to the project.



Key Roles for a Successful Analytics Project

- Figure depicts the various roles and key stakeholders of an analytics project.
- Each plays a critical part in a successful analytics project.
- Although seven roles are listed, fewer or more people can accomplish the work depending on the scope of the project, the organizational structure, and the skills of the participants.
- For example, on a small, versatile team, these seven roles may be fulfilled by only 3 people, but a very large project may require 20 or more people.

Data Analytics Life Cycle



Overview of the Data Analytics Lifecycle

- Figure presents an overview of the Data Analytics Lifecycle that includes six phases.
- Teams commonly learn new things in a phase that cause them to go back and refine the work done in prior phases based on new insights and information that have been uncovered.
- For this reason, Figure is shown as a cycle.
- The circular arrows convey iterative movement between phases until the team members have sufficient information to move to the next phase.
- The callouts include sample questions to ask to help guide whether each of the team members has enough information and has made enough progress to move to the next phase of the process.
- Note that these phases do not represent formal stage gates; rather, they serve as criteria to help test whether it makes sense to stay in the current phase or move to the next.

Phase I Discovery:

- In Phase I, the team learns the business domain, including relevant history such as whether the organization or business unit has attempted similar projects in the past from which they can learn.
- The team assesses the resources available to support the project in terms of people, technology, time, and data.
- Important activities in this phase include framing the business problem as an analytics challenge that can be addressed in subsequent phases and formulating initial hypotheses (IHs) to test and begin learning the data.

Phase 2 Data preparation

- Phase 2 requires the presence of an analytic sandbox, in which the team can work with data and perform analytics for the duration of the project.
- The team needs to execute extract, load, and transform (ELT) or extract, transform and load (ETL) to get data into the sandbox.
- The ELT and ETL are sometimes abbreviated as ETLT.
- Data should be transformed in the ETLT process so the team can work with it and analyze it.
- In this phase, the team also needs to familiarize itself with the data thoroughly and take steps to condition the data.

Phase 3 Model planning

- Phase 3 is model planning, where the team determines the methods, techniques, and workflow it intends to follow for the subsequent model building phase.
- The team explores the data to learn about the relationships between variables and subsequently selects key variables and the most suitable models.

Phase 4 Model building

- In Phase 4, the team develops datasets for testing, training, and production purposes.
- In addition, in this phase the team builds and executes models based on the work done in the model planning phase.
- The team also considers whether its existing tools will suffice for running the models, or if it will need a more robust environment for executing models and workflows (for example, fast hardware and parallel processing, if applicable).

Phase 5 Communicate results

- In Phase 5, the team, in collaboration with major stakeholders, determines if the results of the project are a success or a failure based on the criteria developed in Phase I.
- The team should identify key findings, quantify the business value, and develop a narrative to summarize and convey findings to stakeholders.

Phase 6 Operationalize

- In Phase 6, the team delivers final reports, briefings, code, and technical documents.
- In addition, the team may run a pilot project to implement the models in a production environment.

Phase I: Discovery

- The first phase of the Data Analytics Lifecycle involves discovery.
- In this phase, the data science team must learn and investigate the problem, develop context and understanding, and learn about the data sources needed and available for the project.
- In addition, the team formulates initial hypotheses that can later be tested with data.

Phase I: Discovery

1. Learning the Business Domain
2. Resources
3. Framing the Problem
4. Identifying Key Stakeholders
5. Interviewing the Analytics Sponsor
6. Developing Initial Hypotheses
7. Identifying Potential Data Sources

I. Learning the Business Domain

1. Know the Field:

- It's important for data experts to really understand the area they're working on.

2. Different Types of Data Scientists:

- Data experts can have fancy degrees in math or stats, and they can use their skills in many different areas.

3. Mixing Knowledge:

- Some data experts know a lot about a specific field (like biology) and also know how to use numbers and stats.

I. Learning the Business Domain

4. Plan Early:

- Teams should figure out how much "real-world" knowledge the data folks need right at the start of a project.
- This helps decide who should be on the team and what resources are needed.

5. Success Depends on the Team:

- If the team has the right mix of people who know the subject and those who know the technical stuff, the project is more likely to succeed.

2. Resources

1. Assessment of Resources:

- During the discovery phase, the team needs to evaluate the resources available, including technology, tools, systems, data, and people.

2. Scoping Technology and Tools:

- Consider the tools and technology to be used and the systems needed for later phases to operationalize the models.

3. Analytical Sophistication and Skill Gaps:

- Evaluate the organization's level of analytical sophistication and identify any gaps in tools, technology, and skills.
- Consider what skills and roles may be required in the future for long-term success.

2. Resources

4. Longevity of the Model:

- Consider the skills and roles needed for the model's longevity.
- Assess whether the required expertise exists within the organization or needs to be developed.

5. Inventory of Available Data:

- Take stock of the types of data available for the project.
- Determine if the existing data is sufficient to support the project's goals or if additional data collection, purchase, or transformation is necessary.

2. Resources

6. Strategic Perspective on Data:

- Consider long-term goals and data needs beyond current constraints.
- Determine what can be achieved with existing data and view the project as the first step in a more strategic initiative.

7. Team Composition:

- Ensure the project team has a balanced mix of domain experts, customers, analytic talent, and project management for effectiveness.

8. Evaluation of Time and Skills:

- Assess the time needed for the project and ensure the team possesses the right breadth and depth of skills.

2. Resources

9. Resource Sufficiency:

- Consider if the team has enough resources, including tools, technology, data, and people, to succeed in the project.
- Identify whether additional resources are needed.

10. Negotiation for Resources:

- Negotiate for resources during the project's outset, aligning with scoping goals and objectives.
- This is more effective than negotiating later in the process and ensures proper execution.

11. Proactive Resource Negotiation:

- Project managers and key stakeholders are advised to proactively negotiate for the right resources early in the project rather than waiting until it's underway for better

3. Framing the Problem

1. Critical Role of Problem Framing:

- Framing the problem well is critical to the success of the project.
- Framing is the process of stating the analytics problem to be solved.

2. Documenting the Problem Statement:

- Best practice involves writing down the problem statement and sharing it with key stakeholders to ensure a common understanding among team members.

3. Clarity on Current Situation and Challenges:

- The team needs to communicate the current situation and its challenges, ensuring that team members understand the needs and problems from different perspectives.

3. Framing the Problem

4. Identification of Main Objectives:

- Identify the main objectives of the project in business terms, specifying what needs to be achieved and outlining the necessary steps to meet those needs.

5. Defining Success Criteria:

- Clearly define the objectives and success criteria for the project.
- What does the team aim to achieve, and what will be considered a satisfactory outcome?
- Share these with the project team and align with the project sponsor's expectations.

3. Framing the Problem

6. Establishment of Failure Criteria:

- Equally important is establishing failure criteria to guide the team in understanding when to stop or accept results.
- This prevents continuing unproductive efforts and ensures alignment with project sponsors.

7. Balancing Success and Failure Criteria:

- While focusing on success criteria is common, acknowledging failure criteria is essential to avoid persisting in analyses beyond the point of meaningful insights.

3. Framing the Problem

8. Alignment with Project Sponsors:

- Regularly share the statement of goals, success criteria, and failure criteria with the team to confirm alignment with the expectations of project sponsors.

9. Avoiding Unproductive Effort:

- Establishing clear failure criteria prevents participants from continuing analyses when meaningful insights are unlikely, avoiding unproductive efforts and maintaining alignment with project goals.

4. Identifying Key Stakeholders

1. Stakeholder Identification:

- Identify key stakeholders involved in or affected by the project.
- This includes those who benefit from the project or are significantly impacted by its outcomes.

2. Discussion on Success Criteria and Risks:

- Engage in discussions with stakeholders to identify success criteria and key risks associated with the project.
- Understand the expectations of each stakeholder and the criteria they will use to judge the project's success.

4. Identifying Key Stakeholders

3. Domain Exploration with Stakeholders:

- During stakeholder interviews, gather information about the domain area and relevant history from similar analytics projects.
- Identify specific results each stakeholder desires from the project.

4. Articulating Pain Points (Challenges, issues):

- Clearly articulate the pain points that led to initiating the analytics project.
- Understanding these pain points is crucial for addressing them effectively as the project progresses.

4. Identifying Key Stakeholders

5. Setting Clear Expectations:

- Depending on the number of stakeholders, outline the expected level of activity and participation from each stakeholder.
- This helps set clear expectations, preventing delays in approvals and ensuring smooth progress in the analytical process.

5. Interviewing the Analytics Sponsor

1. Set Clear Goals:

- Make sure everyone knows what the project aims to achieve.
- Ask the project sponsor about their vision, goals, and any specific requirements they have.

2. Ask Detailed Questions:

- Dive deep into the sponsor's understanding of the project.
- Use open-ended questions to uncover important details, challenges, and what they think will make the project successful.

5. Interviewing the Analytics Sponsor

3. Understand Expectations:

- Clearly define timelines, budget, and key milestones with the sponsor. Know how they will measure success.
- This helps in managing everyone's expectations and ensures the project aligns with what the sponsor wants.

Tips for Interviewing Project Sponsors

1. Prepare for the interview; draft questions, and review with colleagues.
2. Use open-ended questions; avoid asking leading questions.
3. Probe for details and pose follow-up questions.
4. Avoid filling every silence in the conversation; give the other person time to think.
5. Let the sponsors express their ideas and ask clarifying questions, such as “Why? Is that correct? Is this idea on target? Is there anything else?”
1. Use active listening techniques; repeat back what was heard to make sure the team heard it correctly, or reframe what was said.

Tips for Interviewing Project Sponsors

7. Try to avoid expressing the team's opinions, which can introduce bias; instead, focus on listening.
8. Be mindful of the body language of the interviewers and stakeholders; use eye contact where appropriate, and be attentive.
9. Minimize distractions.
10. Document what the team heard, and review it with the sponsors.

Interview Questions for Project Sponsors

1. What business problem is the team trying to solve?
2. What is the desired outcome of the project?
3. What data sources are available?
4. What industry issues may impact the analysis?
5. What timelines need to be considered?
6. Who could provide insight into the project?
7. Who has final decision-making authority on the project?

Interview Questions for Project Sponsors

8. How will the focus and scope of the problem change if the following dimensions change:

- Time: Analyzing 1 year or 10 years' worth of data?
- People: Assess impact of changes in resources on project timeline.
- Risk: Conservative to aggressive
- Resources: None to unlimited (tools, technology, systems)
- Size and attributes of data: Including internal and external data sources

6. Developing Initial Hypotheses

1. IH Development for Testing:

- The development of Initial Hypotheses (IHs) is a crucial aspect of the discovery phase.
- IHs are ideas that the team can test with data, forming the basis for analytical tests in later project phases.

2. Creativity in Hypothesis Generation:

- It is recommended to generate a few primary hypotheses for testing initially and then creatively develop several more.
- This approach ensures a diverse set of hypotheses, allowing the team to explore various angles and possibilities.

6. Developing Initial Hypotheses

3. Statistical Perspective and Analytical Tests:

- IHs are meant to be tested experimentally, and statistical perspective is essential for hypothesis testing.

4. Stakeholder and Expert Involvement:

- The process involves gathering and assessing hypotheses not only from the team but also from stakeholders and domain experts.
- These individuals, familiar with the domain, offer valuable perspectives on the problem and potential solutions, enriching the pool of ideas to test.

6. Developing Initial Hypotheses

5. Expanding Project Scope and Addressing Stakeholder Interests:

- The IH development process provides opportunities to illuminate operating assumptions held by stakeholders.
- Additionally, it allows the team to explore adjacent spaces in the project scope where it makes sense.
- By addressing the most important interests of stakeholders, the team ensures a more comprehensive and impactful project outcome.

7. Identifying Potential Data Sources

The team should perform five main activities during this step of the discovery phase:

1. Identify data sources
2. Capture aggregate data sources
3. Review the raw data
4. Evaluate the data structures and tools needed
5. Scope the sort of data infrastructure needed for this type of problem

7. Identifying Potential Data Sources

1. Identify data sources:

- Make a list of candidate data sources the team may need to test the initial hypotheses outlined in this phase.
- Make an inventory of the datasets currently available and those that can be purchased or otherwise acquired for the tests the team wants to perform.

2. Capture aggregate data sources:

- This is for previewing the data and providing high-level understanding.
- It enables the team to gain a quick overview of the data and perform further exploration on specific areas.
- It also points the team to possible areas of interest within the data.

7. Identifying Potential Data Sources

3. Review the raw data:

- Obtain preliminary data from initial data feeds.
- Begin understanding the interdependencies among the data attributes, and become familiar with the content of the data, its quality, and its limitations.

4. Evaluate the data structures and tools needed:

- The data type and structure dictate which tools the team can use to analyze the data.
- This evaluation gets the team thinking about which technologies may be good candidates for the project and how to start getting access to these tools.

7. Identifying Potential Data Sources

5. Scope the sort of data infrastructure needed for this type of problem:

- In addition to the tools needed, the data influences the kind of infrastructure that's required, such as disk storage and network capacity.

Phase 2: Data Preparation

- In the second phase of the Data Analytics Lifecycle
- Data preparation involves exploring, preprocessing, and conditioning data.
- An analytics sandbox is created to explore data separately from the production environment.
- ETLT process (Extract, Transform, Load, and Transform) is used to bring data into the sandbox.
- Understanding the data in detail is crucial for project success.
- Data must be conditioned and transformed for subsequent analysis.

Phase 2: Data Preparation

- Data visualizations are used to understand trends, outliers, and relationships.
- Data preparation is labor-intensive, often taking up at least 50% of project time.
- Insufficient data quality can hinder project progress.
- Teams tend to underestimate the importance of data preparation.
- An iterative approach is essential for successful data preparation.

Phase 2: Data Preparation

1. Preparing the Analytic Sandbox
2. Performing ETLT
3. Learning About the Data
4. Data Conditioning
5. Survey and Visualize
6. Common Tools for the Data Preparation Phase

Key Points on the First Subphase of Data Preparation: Preparing the Analytic Sandbox

1. Purpose of the Analytic Sandbox:

- Provides a dedicated workspace for the team to explore data without affecting live production databases.
- Essential for maintaining control over critical databases, like those used for financial reporting.

2. Data Variety in the Analytic Sandbox:

- Best practice is to collect diverse data types, including summary-level aggregated data, structured data, raw data feeds, and unstructured text data.
- Important for Big Data analytics projects requiring high volumes and varieties of data.

Key Points on the First Subphase of Data Preparation: Preparing the Analytic Sandbox

3. Differing Approaches with IT:

- IT often provides limited access to specific data segments, focusing on the minimum required for a given purpose.
- Data science teams, on the other hand, prefer access to a wide range of data for both purpose-driven analyses and experimentation.

4. Collaboration between Data Science and IT:

- Crucial for the data science team to collaborate with IT, clarifying objectives and aligning goals.
- Justification for an analytics sandbox should be provided, emphasizing its separation from traditional IT-governed data warehouses.

Key Points on the First Subphase of Data Preparation: Preparing the Analytic Sandbox

5. Balancing Needs:

- Balancing the needs of data science teams and IT requires positive working relationships and clear communication with data owners.

6. Benefits of an Analytic Sandbox:

- Enables more ambitious data science projects beyond traditional analysis and Business Intelligence.
- Facilitates advanced predictive analytics.

Key Points on the First Subphase of Data Preparation: Preparing the Analytic Sandbox

7. Sandbox Size and Planning:

- Sandbox size varies based on the project, but planning for 5–10 times the size of original datasets is a good rule.
- May contain raw data, aggregated data, and less commonly used data types.

8. Progress in Industry Adoption:

- While the concept is relatively new, companies are making progress in offering sandboxes that satisfy both data science teams and IT groups.

2. Performing ETLT

1. Bandwidth and Network Connections:

- Ensure the analytic sandbox has ample bandwidth and reliable network connections to the underlying data sources for uninterrupted read and write.

2. ETL vs. ELT Approach:

- Analytic sandbox adopts an Extract, Load, and then Transform (ELT) approach.
- Preserving raw data before transformation is emphasized, providing value in analyzing outliers and hidden patterns.

2. Performing ETLT

3. Importance of Preserving Raw Data:

- Preserving raw data is crucial for analyses, such as fraud detection, where outliers represent valuable information.
- ELT allows access to clean data for analysis and preserves the original data for uncovering hidden insights.

4. Data Movement Parallelization:

- Consider parallelizing data movement, often referred to as Big ETL, using technologies like Hadoop or MapReduce.
- These technologies enable rapid parallel data ingest and processing, aiding in data loading and analysis.

2. Performing ETLT

5. Transformations and Data Quality Assessment:

- Determine transformations needed and assess data quality before moving data into the sandbox.
- Properly structure datasets for robust analysis in subsequent phases.

6. Inventory and Gap Analysis:

- Conduct an inventory of available data and perform gap analysis.
- Understand which datasets are accessible and identify projects needed for data collection or access to new datasets.

2. Performing ETLT

7. Data Connection Considerations:

- Extracting data from the available sources and determining data connections for raw data, online transaction processing (OLTP) databases, online analytical processing (OLAP) cubes, or other data feeds.

8. APIs for Data Access:

- Application Programming Interfaces (APIs) are increasingly popular for accessing data sources.
- APIs, such as those provided by websites and social networks, offer additional data for projects (e.g., sentiment analysis using Twitter data).
- APIs, like the Twitter API, can enrich datasets by providing additional relevant data for analysis.

3. Learning About the Data

- A critical aspect of a data science project is to become familiar with the data itself.
- Spending time to learn the nuances of the datasets provides context to understand what constitutes a reasonable value and expected output versus what is a surprising finding.
- In addition, it is important to catalog the data sources that the team has access to and identify additional data sources that the team can leverage but perhaps does not have access to today.
- Some of the activities in this step may overlap with the initial investigation of the datasets that occur in the discovery phase.

3. Learning About the Data

- Doing this activity accomplishes several goals.
 - Clarifies the data that the data science team has access to at the start of the project
 - Highlights gaps by identifying datasets within an organization that the team may find useful but may not be accessible to the team today. As a consequence, this activity can trigger a project to begin building relationships with the data owners and finding ways to share data in appropriate ways. In addition, this activity may provide an impetus to begin collecting new data that benefits the organization or a specific long-term project.
 - Identifies datasets outside the organization that may be useful to obtain, through open APIs, data sharing, or purchasing data to supplement already existing datasets.

3. Learning About the Data

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 - Identifies datasets outside the organization that may be useful to obtain, through open APIs, data sharing, or purchasing data to supplement already existing datasets.

Sample Dataset Inventory: Table demonstrates one way to organize this type of data inventory.

Dataset	Data Available and Accessible	Data Available, but not Accessible	Data to Collect	Data to Obtain from Third Party Sources
Products shipped	•			
Product Financials		•		
Product Call Center Data		•		
Live Product Feedback Surveys			•	
Product Sentiment from Social Media				•

4. Data Conditioning

1. Data Conditioning Overview:

- Data conditioning is a critical step in the Data Analytics Lifecycle, involving the cleaning, normalization, and transformation of data to prepare it for analysis.

2. Complex Operations Involved:

- The process includes complex steps such as joining or merging datasets to ensure they are in an optimal state for analysis in subsequent phases.

4. Data Conditioning

3. Preprocessing for Analysis:

- Often considered a preprocessing step, data conditioning is performed before developing models, making it a crucial phase in the overall data analysis process.

4. Involvement of Various Roles:

- While traditionally seen as the responsibility of IT, data owners, DBAs, or data engineers, it is essential to involve data scientists in the data conditioning phase.

4. Data Conditioning

5. Decision-Making in Data Conditioning:

- Decisions made during data conditioning, such as selecting aspects of datasets for analysis, can impact subsequent phases, emphasizing the need for collaboration among team members.

6. Thoughtful Data Retention:

- Thoughtful decisions about which data to keep and discard are crucial, as discarding too much data early in the process may lead to retracing steps in the analysis.

5. Survey and Visualize

1. Data Overview with Visualization:

- Data visualization tools offer a valuable step after collecting datasets, providing a high-level overview of the data.

2. Quick Understanding of Data Characteristics:

- Visualization allows for the quick understanding of data characteristics and patterns, aiding in the identification of high-level insights.

5. Survey and Visualize

3. Data Quality Assessment:

- Visualization can be used to examine data quality, identifying unexpected values or indicators of dirty data.

4. Detection of Skewness:

- Skewness, such as data heavily shifted toward one value, can be easily detected through visualization.

5. Survey and Visualize

5. Practical Benefits:

- Enables users to locate areas of interest, zoom in for more detailed information, and obtain specific details about a particular area of the data.

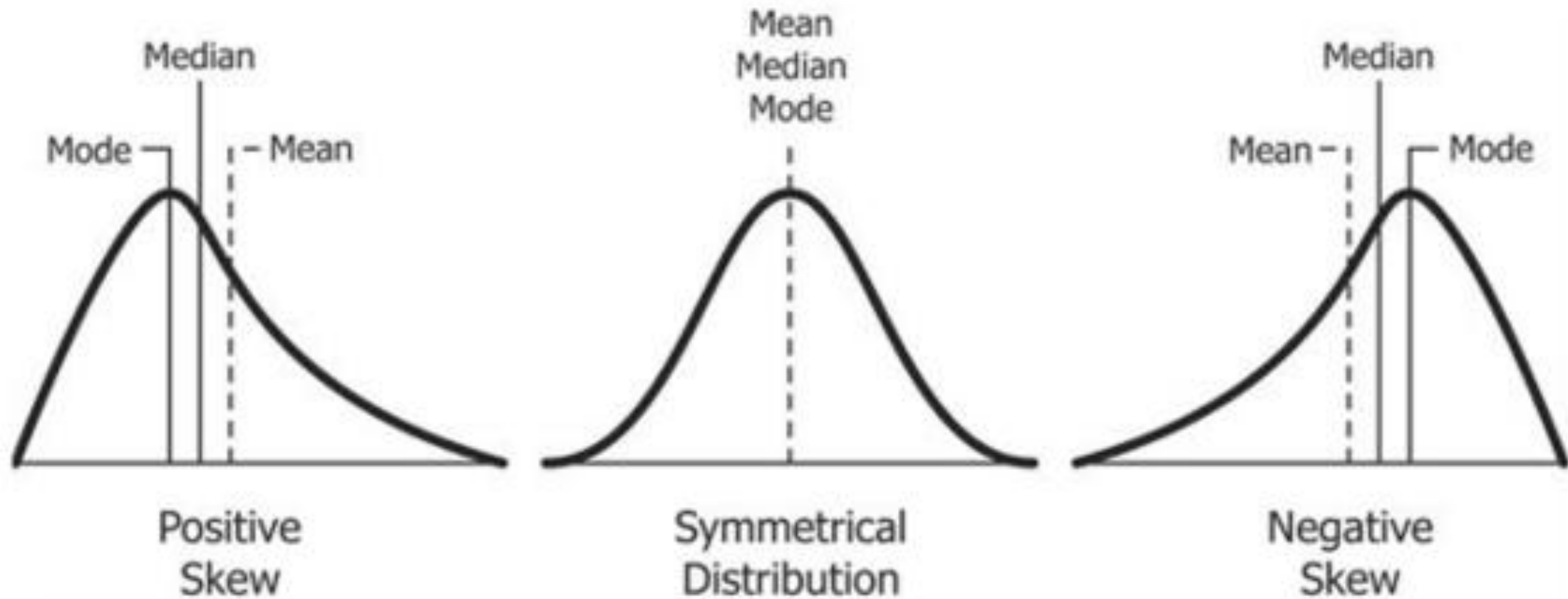
6. Efficient Information Retrieval:

- This approach efficiently provides a high-level view and detailed information about a dataset in a relatively short time.

Skewness

- Skewness is the measure of how much the probability distribution of a random variable deviates from the normal distribution.
- The normal distribution is the probability distribution without any skewness.
- You can look at the image, which shows symmetrical distribution that's a normal distribution, and you can see that it is symmetrical on both sides of the dashed line.
- Apart from this, there are two types of skewness:
 - Positive Skewness
 - Negative Skewness

Skewness



6. Common Tools for the Data Preparation Phase

1. Hadoop:

- Hadoop can perform massively parallel ingest and custom analysis for web traffic parsing, GPS location analytics, genomic analysis, and combining of massive unstructured data feeds from multiple sources.

2. Alpine Miner:

- Alpine Miner provides a graphical user interface (GUI) for creating analytic workflows, including data manipulations and a series of analytic events such as staged data-mining techniques on Postgres SQL(RDBMS) and other Big Data sources.

6. Common Tools for the Data Preparation Phase

3. OpenRefine (formerly called Google Refine):

- It is “a free, open source, powerful tool for working with messy data.”
- It is a popular GUI-based tool for performing data transformations, and it's one of the most robust free tools currently available.

6. Common Tools for the Data Preparation Phase

4. Data Wrangler:

- Similar to OpenRefine, Data Wrangler is an interactive tool for data cleaning and transformation.
- Wrangler was developed at Stanford University and can be used to perform many transformations on a given dataset.
- In addition, data transformation outputs can be put into Java or Python.
- The advantage of this feature is that a subset of the data can be manipulated in Wrangler via its GUI, and then the same operations can be written out as Java or Python code to be executed against the full, larger dataset offline in a local analytic sandbox.

Phase 3: Model Planning

- In Phase 3, the data science team identifies candidate models to apply to the data for clustering, classifying, or finding relationships in the data depending on the goal of the project.
- It is during this phase that the team refers to the hypotheses developed in Phase I, when they first became acquainted with the data and understanding the business problems or domain area.
- These hypotheses help the team frame the analytics to execute in Phase 4 and select the right methods to achieve its objectives.

Phase 3: Model Planning

- Some of the activities to consider in this phase include the following:
 - Assess the structure of the datasets. The structure of the datasets is one factor that dictates the tools and analytical techniques for the next phase. Depending on whether the team plans to analyze textual data or transactional data, for example, different tools and approaches are required.
 - Ensure that the analytical techniques enable the team to meet the business objectives and accept or reject the working hypotheses.

Phase 3: Model Planning

- Determine if the situation warrants a single model or a series of techniques as part of a larger analytic workflow. A few example models include association rules and logistic regression. Other tools, such as Alpine Miner, enable users to set up a series of steps and analyses and can serve as a front-end user interface (UI) for manipulating Big Data sources in PostgreSQL.

Phase 3: Model Planning

- In addition to the considerations just listed, it is useful to research and understand how other analysts generally approach a specific kind of problem.
- Given the kind of data and resources that are available, evaluate whether similar, existing approaches will work or if the team will need to create something new.
- Many times teams can get ideas from analogous problems that other people have solved in different industry verticals or domain areas.

Phase 3: Model Planning

- Table summarizes the results of an exercise of this type, involving several domain areas and the types of models previously used in a classification type of problem after conducting research on churn models in multiple industry verticals.
- Performing this sort of diligence gives the team ideas of how others have solved similar problems and presents the team with a list of candidate models to try as part of the model planning phase.

Table: Research on Model Planning in Industry Verticals

Market Sector	Analytic Techniques/Methods Used
Consumer Packaged Goods	Multiple linear regression, automatic relevance determination (ARD), and decision tree
Retail Banking	Multiple regression
Retail Business	Logistic regression, ARD, decision tree
Wireless Telecom	Neural network, decision tree, hierarchical neurofuzzy systems, rule evolver, logistic regression

Phase 3: Model Planning

1. Data Exploration and Variable Selection
2. Model Selection
3. Common Tools for the Model Planning Phase

Data Exploration and Variable Selection

- The data exploration phase is essential for understanding variable relationships and informing the selection of variables and methods in subsequent phases.
- Stakeholders and subject matter experts play a key role by providing insights and hypotheses, guiding the exploration with their understanding of the problem domain.
- Data scientists approach the exploration with an unbiased mindset, questioning assumptions and rigorously testing initial ideas presented by stakeholders.
- Utilizing tools for data visualizations enables the team to preview the data and assess relationships between variables at a high level.

Data Exploration and Variable Selection

- The focus is on capturing essential predictors and variables, iterating through testing to prioritize the most influential ones for the intended analyses.
- In regression analysis planning, identifying candidate predictors and prioritizing variables with strong relationships to outcomes is crucial.
- Vigilance is maintained for modeling challenges such as serial correlation and multicollinearity, with consideration given to reframing problems to avoid complications.
- Recognition of diverse modeling objectives involves understanding when "black box prediction" suffices or when explanatory power is essential for causal understanding.

Data Exploration and Variable Selection

- The emphasis on explanatory power is particularly relevant in cases where a deeper understanding of causal relationships is required, and stress testing models is done to ensure robustness.
- The iterative nature of the process is highlighted, with an emphasis on efficient and effective exploration to streamline subsequent phases in the data analytics lifecycle.

Model Selection

- In the model selection subphase, the primary objective of the team is to choose an analytical technique or a shortlist of candidate techniques aligned with the project's end goal.
- A model is broadly defined as an abstraction from reality, attempting to emulate real-world behavior with a set of rules and conditions, particularly in the realms of machine learning and data mining.
- Various types of models, such as classification, association rules, and clustering, are considered, with the team narrowing down the options based on the specific problem at hand.

Model Selection

- Dealing with Big Data requires an additional consideration, where the team decides whether the chosen techniques are best suited for structured data, unstructured data, or a hybrid approach.
- It is emphasized that the team should carefully identify and document the modeling assumptions made during the selection and construction of preliminary models.
- Initial models are typically created using statistical software packages like R, SAS, or Matlab, designed for data mining and machine learning algorithms. However, limitations may arise when applying these models to very large datasets, common in Big Data scenarios.

Model Selection

- Considering these limitations, the team may contemplate redesigning algorithms to run in the database itself during the pilot phase mentioned in Phase 6.
- Transitioning to the model building phase is dependent on having a clear idea of the type of model to try, gaining sufficient knowledge to refine the analytics plan, and establishing a general methodology for the chosen analytical model.
- Advancing from this phase necessitates a solid understanding of variables and techniques, as well as a description or diagram outlining the analytic workflow.

Common Tools for the Model Planning Phase

■ R:

- R has a complete set of modeling capabilities and provides a good environment for building interpretive models with high-quality code.
- In addition, it has the ability to interface with databases via an ODBC connection and execute statistical tests and analyses against Big Data via an open source connection.
- These two factors make R well suited to performing statistical tests and analytics on Big Data.
- R contains nearly 5,000 packages for data analysis and graphical representation.

Common Tools for the Model Planning Phase

- **SQL Analysis services:**

- SQL Analysis services can perform in-database analytics of common data mining functions, involved aggregations, and basic predictive models.

- **SAS/ACCESS:**

- SAS/ACCESS provides integration between SAS (Statistical Analysis System) and the analytics sandbox via multiple data connectors such as ODBC, JDBC, and OLE DB (Object Linking and Embedding Database).
- SAS itself is generally used on file extracts, but with SAS/ACCESS, users can connect to relational databases (such as Oracle or Teradata) and data warehouse appliances (such as Greenplum or Aster), files, and enterprise applications (such as SAP and Salesforce.com).

Phase 4: Model Building

- In Phase 4, the data science team needs to develop datasets for training, testing, and production purposes.
- These datasets enable the data scientist to develop the analytical model and train it (“training data”), while holding aside some of the data (“hold-out data” or “test data”) for testing the model.
- During this process, it is critical to ensure that the training and test datasets are sufficiently robust for the model and analytical techniques.
- A simple way to think of these datasets is to view the training dataset for conducting the initial experiments and the test sets for validating an approach once the initial experiments and models have been run.

Phase 4: Model Building

- In the model building phase, an analytical model is developed and fit on the training data and evaluated (scored) against the test data.
- The phases of model planning and model building can overlap quite a bit, and in practice one can iterate back and forth between the two phases for a while before settling on a final model.

Common Tools for the Model Building Phase: Commercial Tools

■ **SAS Enterprise Miner:**

- SAS Enterprise Miner allows users to run predictive and descriptive models based on large volumes of data from across the enterprise.
- It interoperates with other large data stores, has many partnerships, and is built for enterprise-level computing and analytics.

■ **SPSS Modeler:**

- SPSS Modeler (provided by IBM and now called IBM SPSS Modeler) offers methods to explore and analyze data through a GUI.

Common Tools for the Model Building Phase: Commercial Tools

- **Matlab:**

- Matlab provides a high-level language for performing a variety of data analytics, algorithms, and data exploration.

- **Alpine Miner:**

- Alpine Miner provides a GUI front end for users to develop analytic workflows and interact with Big Data tools and platforms on the back end.

- **STATISTICA and Mathematica:**

- STATISTICA and Mathematica are also popular and well-regarded data mining and analytics tools.

Common Tools for the Model Building Phase: Free or Open Source tools

■ **R and PL/R:**

- R was described earlier in the model planning phase, and PL/R is a procedural language for PostgreSQL with R.
- Using this approach means that R commands can be executed in database.
- This technique provides higher performance and is more scalable than running R in memory.

■ **Octave:**

- A free software programming language for computational modeling, has some of the functionality of Matlab.
- Because it is freely available, Octave is used in major universities when teaching machine learning.

Common Tools for the Model Building Phase: Free or Open Source tools

■ **WEKA:**

- WEKA is a free data mining software package with an analytic workbench.
- The functions created in WEKA can be executed within Java code.

■ **Python:**

- Python is a programming language that provides toolkits for machine learning and analysis, such as scikit-learn, numpy, scipy, pandas, and related data visualization using matplotlib.

■ **SQL:**

- SQL in-database implementations, such as MADlib , provide an alternative to in-memory desktop analytical tools.
- MADlib provides an open-source machine learning library of algorithms that can be executed in-database, for PostgreSQL or Greenplum.

Phase 5: Communicate Results

- After executing the model, the team needs to compare the outcomes of the modeling to the criteria established for success and failure.
- In Phase 5, the team considers how best to articulate the findings and outcomes to the various team members and stakeholders, taking into caution, assumptions, and any limitations of the results.
- Because the presentation is often circulated within an organization, it is critical to articulate the results properly and position the findings in a way that is appropriate for the audience.

Phase 5: Communicate Results

- As part of Phase 5, the team needs to determine if it succeeded or failed in its objectives.
- Many times people do not want to admit to failing, but in this instance failure should not be considered as a true failure, but rather as a failure of the data to accept or reject a given hypothesis adequately.

Phase 6: Operationalize

- In the final phase, the team focuses on communicating project benefits broadly and initiating a pilot project to deploy the work in a controlled manner before a full-scale enterprise rollout.
- Phase 6 represents the first deployment of new analytical methods or models in a production environment, involving a pilot project to manage risks, learn about performance constraints, and make necessary adjustments before wider implementation.
- During the pilot, considerations include executing the algorithm in the database for faster and more efficient run time, especially with larger datasets, and scoping the effort to test the model in a live setting for a discrete set of products or a single line of business.

Phase 6: Operationalize

- This phase may involve new team members, particularly engineers responsible for the production environment, addressing technical issues and ensuring smooth integration with related business processes.
- Operationalizing includes creating mechanisms for ongoing monitoring of model accuracy, retraining the model if accuracy degrades, and designing alerts for out-of-bounds situations when inputs exceed the model's training range.